AIDCF-018-SUP Supply Chain, OEM and Ecosystem Assurance Domain Guide

Document control

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Revision history

Version	Date	Summary
1.0	2026-10-10	First issue of the SUP domain guide, aligned with the rewritten SUP controls in AIDCF-007A and the per-control gate mapping rule.

Status of this guide

This guide explains how to apply the 24 SUP controls of AIDCF-007A. The control records are normative and are held in the AIDCF-007A registry; where this guide and a control record differ, the control record governs and the difference shall be raised as a baseline issue. Requirements numbered DG-SUP-REQ-NNN are normative for assessments that claim conformance to AIDCF-018-SUP. Other text is informative.

1 Purpose and scope

1.1 Purpose

The SUP domain establishes whether the organisations, products, firmware, spares, logistics and remote-support arrangements on which the AI Capability Instance depends are trustworthy, contractually secured, traceable and recoverable. Where DEP asks "what do we depend on?", SUP asks "can we rely on the parties and products we depend on, and can we get what we need from them when it matters?".

1.2 In scope

- Supplier due diligence: technical capability, financial and operational resilience, security and quality posture, sub-tier visibility (SUP-01).

- Contract and service obligations: service levels, evidence rights, incident and vulnerability notification, exit and transition rights (SUP-02).
- Hardware and component provenance: approved sourcing, counterfeit prevention, chain of custody, component traceability (SUP-03).
- Software, firmware and update supply: release provenance, signature validation, compatibility assurance and staged rollout, vulnerability response (SUP-04).
- Spares, logistics and obsolescence: critical spares, long-lead equipment lead times, logistics and specialist-labour continuity, end-of-life planning (SUP-05).
- Remote support and third-party access: named identities, activation and approval, session monitoring, revocation and review (SUP-06).

The supply base in scope includes accelerator and rack-scale system OEMs and integrators; NIC/DPU, switch and optics suppliers; storage OEMs; MV/LV switchgear, transformer, UPS, battery, generator and fuel suppliers; chiller, dry cooler, cooling tower, CDU and pump suppliers; BMS/EPMS and controls integrators; maintenance contractors; logistics providers; and managed-service and remote-support providers.

1.3 Out of scope

- Identifying that a dependency on a supplier or provider exists, and analysing its concentration (DEP-03, DEP-04).
- Platform security baselines, vulnerability management and patching of the owner's own systems (SEC-04), and the owner's own privileged access model (SEC-01). SUP covers the supplier side of these.
- Physical delivery, equipment movement and media custody within the site (PHY-05).
- Node rebuild, spares and RMA handling as part of compute operations (CMP-06); SUP sets the supply strategy that CMP-06 executes.

1.4 Boundaries with other domains

Topic	SUP role	Other domain role
OEM dependency	Assures the supplier and contract (SUP-01, SUP-02)	DEP-03 records the dependency (MCF-DEP-011)
Supplier concentration	Provides sub-tier facts (MCF-SUP-004)	DEP-04 analyses concentration (MCF-DEP-014)
Firmware	Assures supply, provenance, signature and rollout (SUP-04)	CMP-04 assures the compute firmware baseline and upgrade testing; SEC-04 assures secure baseline and patching
Remote access	Assures third-party identity and session lifecycle (SUP-06)	SEC-01/SEC-03 assure identity, privileged and remote access architecture
Spares	Sets spares and lead-time strategy (SUP-05)	CMP-06 executes compute spares and RMA; POW/THM restoration uses spares
Delivery	Assures chain of custody up to receipt (MCF-SUP-011)	PHY-05 assures delivery controls and movement on site

2 Why this domain matters for AI capability

AI infrastructure is built from a small number of highly specialised products made by a small number of suppliers, many of them supply-constrained. Accelerators, high-bandwidth memory, advanced packaging and high-speed optics are allocated rather than freely available; power transformers, MV switchgear, large generators, chillers and CDUs have long and variable manufacturing lead times; specialist labour for HV jointing, commissioning and liquid-cooling work is scarce in many regions. A failure that would be routine in a general-purpose facility (replace the part from stock) can become a long outage when the part is a long-lead item with no strategic spare.

The software supply chain is as material as the hardware. Accelerators, BMCs, NICs/DPUs, switches, storage controllers, UPS, PDUs, CDU controllers and PLCs all run vendor firmware that is updated over the life of the facility. A defective or malicious release, rolled out across all redundant members at once, is a common-mode failure (AIDCF-004 scenario X08) that no amount of physical redundancy prevents. Vendor remote access, often needed for diagnosis of proprietary systems, gives third parties consequential authority over facility and AI infrastructure and is directly relevant to CG-15.

Finally, supplier contracts determine what evidence the owner can obtain. Where a colocation provider, OEM or integrator will not disclose test records, sub-tier sources or firmware component inventories, the owner cannot reach the evidence floors for dependent controls in other domains, and the result is UNKNOWN rather than CONFORMING.

3 Reference patterns and variants

Pattern (AIDCF-013)	Dominant SUP concerns	Controls most exercised
Single-site training facility	Long-lead spares for site-unique electrical and cooling plant; accelerator RMA turnaround	MCF-SUP-017, MCF-SUP-018, MCF-SUP-005
Hyperscale campus	Standardised fleets with shared firmware lineage; framework agreements and slot reservations	MCF-SUP-015, MCF-SUP-018, MCF-SUP-004
Colocation deployment	Evidence rights against the provider; provider's maintenance contractors	MCF-SUP-006, MCF-SUP-005, MCF-SUP-021
Liquid-cooled high-density cluster	CDU and cold-plate OEM support; coolant supply; proprietary control logic	MCF-SUP-008, MCF-SUP-013, MCF-SUP-017
Sovereign AI environment	Supplier jurisdiction, export controls, sub-tier disclosure, support personnel clearance	MCF-SUP-002, MCF-SUP-004, MCF-SUP-021
Edge inference estate	Many sites with remote vendor support; logistics to dispersed sites	MCF-SUP-019, MCF-SUP-022
Brownfield retrofit	Obsolete controllers and UPS models; broker-sourced parts	MCF-SUP-020, MCF-SUP-009, MCF-SUP-010
Modular facility	Integrator-built skids with embedded controls; factory acceptance testing	MCF-SUP-001, MCF-SUP-013, MCF-SUP-011

Variants. In an *integrator-delivered* model (rack-scale or skid-level systems assembled and tested by an integrator), SUP-01 and SUP-03 controls are applied to the integrator and, through sub-tier visibility, to the OEMs behind it. In a *managed-service* model, SUP-06 applies to the managed-service provider's staff as third parties even when they are permanently on site.

4 Required functions and capability linkage

Required Function (assurance view)	Linked Service Outcomes	Controls
Supplier fitness: critical suppliers can deliver, support and survive	Availability, growth and recovery outcomes	MCF-SUP-001 to MCF-SUP-004
Contractual reliance: obligations, evidence, notification and exit are secured	All outcomes (through evidence and support)	MCF-SUP-005 to MCF-SUP-008
Hardware integrity: installed components are genuine, untampered and traceable	Integrity, availability and life-safety outcomes	MCF-SUP-009 to MCF-SUP-012
Firmware and software integrity: updates are authentic, compatible and safely deployed	Integrity and availability outcomes	MCF-SUP-013 to MCF-SUP-016

Required Function (assurance view)	Linked Service Outcomes	Controls
Restoration supply: spares, long-lead equipment, logistics and labour are available within tolerance	Recovery outcomes	MCF-SUP-017 to MCF-SUP-020
Bounded third-party authority: vendor access is identified, approved, monitored and revoked	Integrity and availability outcomes; CG-15	MCF-SUP-021 to MCF-SUP-024

In the AIDCF-014 graph, a supplier or provider appears as the provider end of a Dependency (DEPENDS_ON from the consuming system or function). Contract evidence attaches through HAS_EVIDENCE_REQUIREMENT and EVIDENCED_BY; firmware releases attach to AssetInstances; and third-party access paths are Interfaces CONTROLLED_BY the owner's access gateway.

5 Dependencies, fault domains and common-mode concerns

Common mode	How it arises	Control
Single firmware lineage across redundant members	Standardisation plus a single rollout pipeline	MCF-SUP-015 (fault-domain-aligned waves; redundant members never in one wave)
Shared sub-tier source behind "diverse" suppliers	Two UPS brands with one battery-cell maker; two optics vendors with one laser source	MCF-SUP-004
One integrator or contractor across systems	Same contractor maintains all generators or both UPS systems	MCF-SUP-001; DEP-02 shared human dependency
One logistics route or port for heavy equipment	Transformers and generators delivered through one route or crossing	MCF-SUP-019
One vendor remote-access path to many sites	A vendor's support platform with standing access to the estate	MCF-SUP-022
One spares location	All strategic spares in the same building as the equipment they protect	MCF-SUP-017

The human/operational and external fault-domain categories of AIDCF-003 §43 ("team, shift, procedure, supplier or authority concentration"; "utility, carrier, water source, fuel supply, vendor or geographic region") are where SUP findings are recorded.

DG-SUP-REQ-001 The assessor shall confirm that each firmware rollout sampled under MCF-SUP-015 was planned against the RES-02 fault-domain model current at the time of the rollout, and shall record as a Finding any wave that included more than one member of a redundant group.

6 Threats, hazards and failure scenarios with propagation paths

Scenario	Initiating condition	Propagation path to Service Outcome	SUP controls that interrupt the path
X08 Firmware rollout common-mode failure	Defective NIC or switch firmware release	Deployed to all spine switches in one change → fabric partition → collective operations fail → training jobs fail across the cluster	MCF-SUP-015 staged waves; MCF-SUP-014 signature validation; MCF-SUP-013 provenance
X12 Long-lead spare after rare failure	Internal fault in a site transformer or MV switchgear section	No strategic spare → replacement lead time far exceeds tolerance → prolonged operation at reduced redundancy or reduced capacity → availability and growth outcomes missed	MCF-SUP-017, MCF-SUP-018, MCF-SUP-019

Scenario	Initiating condition	Propagation path to Service Outcome	SUP controls that interrupt the path
X18 Loss of remote OEM support	OEM support platform unavailable, or supplier withdraws support	Diagnosis of a proprietary storage or fabric fault delayed → recovery beyond tolerance	MCF-SUP-005, MCF-SUP-008, MCF-SUP-020
Supply-chain compromise (AIDCF-004 threat event)	Modified BMC firmware or tampered network device introduced in transit	Persistent foothold on management plane → manipulation of power capping or cooling setpoints	MCF-SUP-011, MCF-SUP-014, MCF-SUP-010
Counterfeit protection device	Counterfeit MCCB or relay installed	Fails to clear a fault → arc-flash hazard and wider outage than the protection study assumed	MCF-SUP-009, MCF-SUP-010 (CG-04 SUPPORTING)
Vendor access abuse or compromise	Standing shared vendor account on CDU controllers	Unauthorised setpoint change across loops → thermal excursion	MCF-SUP-021, MCF-SUP-022, MCF-SUP-023
Supplier distress or export restriction	Supplier insolvency, plant outage or export-control change	Accelerator deliveries or spares stop → expansion stalls; RMA backlog grows	MCF-SUP-002, MCF-SUP-004, MCF-SUP-018

7 Safeguards

Safeguard	Function	Controls
Pre-award capability and posture assessment	Prevents reliance on suppliers unable to deliver or support	MCF-SUP-001, MCF-SUP-003
Supplier monitoring	Detects distress, allocation and export exposure early	MCF-SUP-002
Measurable service levels tied to tolerances	Aligns support with time-to-impact	MCF-SUP-005
Evidence rights	Makes supplier-held evidence obtainable	MCF-SUP-006
Approved sourcing, receipt inspection and custody	Excludes counterfeit and tampered parts	MCF-SUP-009 to MCF-SUP-011
Serial-level traceability	Scopes recalls and advisories quickly	MCF-SUP-012
Controlled release repository and signature enforcement	Excludes unauthentic firmware	MCF-SUP-013, MCF-SUP-014
Compatibility matrix and fault-domain-aligned rollout	Contains defective releases	MCF-SUP-015
Strategic spares, lead-time tracking and logistics pre-arrangements	Bounds restoration time	MCF-SUP-017 to MCF-SUP-019
Named, off-by-default, brokered and recorded vendor access	Bounds third-party authority	MCF-SUP-021 to MCF-SUP-024

8 Critical-gate linkage

SUP holds three CRITICAL mappings: MCF-SUP-004 for CG-03, and MCF-SUP-021 and MCF-SUP-022 for CG-15. All other mappings are SUPPORTING; SUP findings frequently drive gate-level evaluation of CG-08, CG-10, CG-14 and CG-15 and supporting evaluation of CG-04 (counterfeit protection devices) and CG-05/CG-06 (spares and logistics).

Gate	Gate name	CRITICAL controls in this domain	SUPPORTING controls in this domain
CG-03	Critical dependency visibility	MCF-SUP-004	MCF-SUP-001, MCF-SUP-002, MCF-SUP-006, MCF-SUP-007, MCF-SUP-012, MCF-SUP-018, MCF-SUP-020
CG-04	Life safety and hazardous energy	none	MCF-SUP-009, MCF-SUP-010
CG-05	Resilient electrical supply	none	MCF-SUP-017, MCF-SUP-019
CG-06	Thermal continuity	none	MCF-SUP-017
CG-08	Compute, storage and network functional integrity	none	MCF-SUP-003, MCF-SUP-009, MCF-SUP-010, MCF-SUP-011, MCF-SUP-012, MCF-SUP-014, MCF-SUP-015
CG-09	Failure containment	none	MCF-SUP-015
CG-10	Recovery readiness	none	MCF-SUP-005, MCF-SUP-008, MCF-SUP-017, MCF-SUP-018, MCF-SUP-019
CG-13	Responsibility and authority	none	MCF-SUP-005, MCF-SUP-021, MCF-SUP-024
CG-14	Change control	none	MCF-SUP-013, MCF-SUP-015, MCF-SUP-022
CG-15	Integrated digital authority over infrastructure	MCF-SUP-021, MCF-SUP-022	MCF-SUP-003, MCF-SUP-007, MCF-SUP-013, MCF-SUP-014, MCF-SUP-016, MCF-SUP-023, MCF-SUP-024

CRITICAL controls in this domain:

- **MCF-SUP-004 Sub-Tier Visibility** — CRITICAL for CG-03; claim type IMPLEMENTATION at AL3, sufficiency floor S4.
- **MCF-SUP-021 Support Identity** — CRITICAL for CG-15; claim type IMPLEMENTATION at AL3, sufficiency floor S4.
- **MCF-SUP-022 Activation and Approval** — CRITICAL for CG-15; claim type OPERATION at AL4, sufficiency floor S4.

Rationale for the CRITICAL choices. Sub-tier visibility (MCF-SUP-004) is CRITICAL for CG-03 because a shared sub-tier source behind nominally diverse suppliers is a material dependency that is invisible without it. Support identity and activation (MCF-SUP-021, MCF-SUP-022) are CRITICAL for CG-15 because standing or shared vendor access to BMS/EPMS, CDU, BMC or cluster-management interfaces directly defeats the bound on digital authority that CG-15 protects. Spares and logistics controls are SUPPORTING for CG-10 rather than CRITICAL because the CG-10 CRITICAL controls in POW, THM, CMP, STO, NET and RES evaluate restoration end-to-end; a spares shortfall is assessed through them.

DG-SUP-REQ-002 Where MCF-SUP-021 or MCF-SUP-022 is assessed, the assessor shall include facility control systems (BMS, EPMS, CDU controllers, UPS and switchgear management interfaces) in the sample population alongside IT management interfaces, and shall not record CONFORMING on a sample drawn from IT systems alone.

9 Capability-profile parameters

All defaults are illustrative and shall be replaced in the capability profile with site-specific values and their basis.

Parameter	Unit	Illustrative default	Basis	Control
Supplier monitoring interval	months	6	Illustrative; quarterly for sole-source suppliers.	MCF-SUP-002
Sub-tier depth for critical classes	tiers	2	Illustrative; deeper where single-source components are known.	MCF-SUP-004
Maximum on-site attendance time for critical electrical/mechanical faults	hours	4	Illustrative; set from DEP-003 time-to-impact and spares held.	MCF-SUP-005
Evidence delivery time on request	working days	10	Illustrative.	MCF-SUP-006
Notification intake test interval	months	6	Illustrative.	MCF-SUP-007
Minimum transition assistance period	months	12	Illustrative.	MCF-SUP-008
Serial verification coverage for safety-critical protection devices	% of units received	100	Illustrative.	MCF-SUP-010
Target time to identify all affected units	hours	24	Illustrative.	MCF-SUP-012
Traceability exercise interval	months	12	Illustrative.	MCF-SUP-012
Maximum share of a redundant group updated per wave	% of group	50 (and never all members)	Illustrative.	MCF-SUP-015
Soak time between waves	hours	24	Illustrative.	MCF-SUP-015
Remediation target for critical, exposed vulnerabilities	days	14	Illustrative; OT may require longer with compensating controls.	MCF-SUP-016
Spares stock review interval	months	6	Illustrative.	MCF-SUP-017
Spare-to-installed ratio for accelerator trays	% of installed	Site-specific (no default)	Derived from observed failure rates, RMA turnaround and the compute availability outcome.	MCF-SUP-017
Lead-time refresh interval	months	6	Illustrative; lead times for electrical equipment can change materially within a year.	MCF-SUP-018
Logistics continuity validation interval	months	24	Illustrative.	MCF-SUP-019
Obsolescence planning horizon	months	24	Illustrative.	MCF-SUP-020
Maximum default session duration	hours	4	Illustrative.	MCF-SUP-022
Emergency activation review time	hours	24	Illustrative.	MCF-SUP-022
Session recording retention	days	90	Illustrative.	MCF-SUP-023
Maximum revocation time after notification	hours	24	Illustrative.	MCF-SUP-024

Parameter	Unit	Illustrative default	Basis	Control
Recertification interval	months	3	Illustrative.	MCF-SUP-024

10 Evidence expectations by claim type

Claim type	AL	Profile tier	Controls in this domain	Sufficiency floors applied
DESIGN	AL2	CP-1	MCF-SUP-001, MCF-SUP-008, MCF-SUP-018	S3
IMPLEMENTATION	AL3	CP-2	MCF-SUP-003, MCF-SUP-004, MCF-SUP-005, MCF-SUP-006, MCF-SUP-009, MCF-SUP-013, MCF-SUP-021	S3, S4
OPERATION	AL4	CP-3	MCF-SUP-002, MCF-SUP-007, MCF-SUP-010, MCF-SUP-011, MCF-SUP-016, MCF-SUP-017, MCF-SUP-020, MCF-SUP-022, MCF-SUP-023, MCF-SUP-024	S4
EFFECTIVENESS	AL5	CP-4	MCF-SUP-012, MCF-SUP-014, MCF-SUP-015	S4
RESILIENCE	AL5	CP-5	MCF-SUP-019	S5

DESIGN claims (MCF-SUP-001, MCF-SUP-008, MCF-SUP-018) rest on E1 records (capability assessments, contract clauses, lead-time registers) and analysis, approved by a competent engineer outside the procurement line. They do not show that the supplier performs.

IMPLEMENTATION claims (for example MCF-SUP-004, MCF-SUP-005, MCF-SUP-009, MCF-SUP-013, MCF-SUP-021) require E2 technical-state evidence: purchasing-system configuration, repository inventories reconciled to device-reported versions, and account exports from the systems themselves. A supplier's statement that it uses named accounts is not evidence of the as-configured state.

OPERATION claims (for example MCF-SUP-010, MCF-SUP-016, MCF-SUP-017, MCF-SUP-022 to MCF-SUP-024) require E3 evidence over the period: receipt inspection records, advisory logs, stock reconciliations, gateway session logs and recertification records. The assessor should compute from raw records (session start/end, approval timestamps) rather than rely on summary reports.

EFFECTIVENESS claims (MCF-SUP-012, MCF-SUP-014, MCF-SUP-015) require E4 evidence: a traceability exercise against a defined notice, a rejection test of an unsigned image on non-production equipment, and pre-production compatibility tests with health-gated rollout records.

RESILIENCE claims (MCF-SUP-019) require scenario evidence that logistics and labour can be mobilised within tolerance under a regional-disruption assumption, with cross-domain dependency coverage.

E5 evidence is common in this domain (certificates, audit reports, supplier attestations). AIDCF-008 requires it to be evaluated for scope, method, age, exclusions and transferability; a quality certificate does not cover firmware security, and a supplier's SOC-type report covers only the systems in its stated scope.

11 Assessment guidance

1. **Build the critical-supplier list from DEP.** Use the OEM dependency register (MCF-DEP-011) and supplier concentration analysis (MCF-DEP-014) to select suppliers. Do not sample suppliers by spend.
2. **Read the contract against the control evidence profiles.** For each critical supplier, check whether the contract gives access to the mandatory evidence items of the controls the supplier affects in POW, THM, CMP, STO, NET and RES. Gaps here explain later UNKNOWN results.

3. **Reconcile firmware from the device side.** Pull versions from the devices (Redfish/BMC inventory, switch OS show commands, UPS and CDU controller interfaces, PLC project versions) using read-only methods and reconcile to the repository, not the other way round.
4. **Walk a rollout.** Pick a recent fleet-wide update and reconstruct the waves from deployment records; check them against the fault-domain model and against the redundant-group membership.
5. **Count the spares.** Physical count of a sample of critical spares, with condition check (battery float charge records, storage conditions, firmware level of spare boards), is the only reliable evidence that spares exist and are usable.
6. **Follow a vendor session end to end.** From ticket to approval to activation to recording to deactivation; then check that the vendor account was recertified at the last review.
7. **Treat refusal as data.** A supplier's refusal to disclose sub-tier sources, SBOMs or test records is recorded as an UNKNOWN Record against the affected controls, with the supplier named, and not as NOT_APPLICABLE.

DG-SUP-REQ-003 The assessor shall reconcile deployed firmware and software versions from device-reported data for a sample of each in-scope device class, including at least one facility controller class, when assessing MCF-SUP-013.

DG-SUP-REQ-004 The assessor shall record supplier refusal of evidence required by a control's mandatory items as an UNKNOWN Record of type U2 against each affected control, and shall not substitute a supplier attestation for the mandatory item without recording the confidence ceiling applied.

12 Testing safety

- Signature rejection tests (MCF-SUP-014) are performed only on non-production devices isolated from production management networks. Flashing production devices to test signature enforcement is prohibited.
- Counterfeit testing of protection devices (MCF-SUP-010) is performed on a test bench by competent persons; no device in a live switchboard is tested under this control.
- Account and session reviews (MCF-SUP-021 to MCF-SUP-024) use read-only exports; assessors shall not attempt log-in with vendor or default credentials against production controllers, and shall not revoke access.
- Session termination tests (MCF-SUP-023) use a test session on non-production systems.
- Logistics exercises (MCF-SUP-019) are tabletop or walkthrough; any physical mobilisation trial involving cranes or heavy haulage requires a lift plan and permit under site rules.
- Notification-intake tests (MCF-SUP-007) use clearly marked synthetic notifications agreed with the supplier.

13 UNKNOWN hot-spots

Hot-spot	Typical UNKNOWN type	Typical severity	Resolution route
Sub-tier sources for accelerators, optics, transformers, batteries	U2 / U10	US4 (CG-03 CRITICAL)	Supplier disclosure under NDA; industry mapping; teardown evidence
Firmware SBOM and build provenance for facility controllers	U1 / U6	US3	Evidence rights; OEM release records
Account state on facility controllers that cannot export users	U2	US4 where CG-15 relies on it	Configuration backup review; OEM tool export
Real stock and condition of OEM-held or regional spares	U2	US3	Vendor-managed inventory visibility; audit rights
Transformer and switchgear lead times	U1 / U3	US3	Current quotations; slot reservations

Hot-spot	Typical UNKNOWN type	Typical severity	Resolution route
Supplier business continuity arrangements	U2	US3	Supplier BCP summary; audit
Provider maintenance-contractor competence in colocation	U2	US3	Provider evidence rights; witness

14 Interfaces with other domains

Domain	SUP provides	SUP receives
GOV	Supplier risk acceptance requests	Risk acceptance authority; policy (GOV-02, GOV-03)
CAP	Lead times vs growth schedule	Growth forecast (CAP-04)
DEP	Sub-tier facts; supplier posture	OEM dependency register; concentration analysis (DEP-03, DEP-04)
PHY	Custody to receipt	Delivery and movement controls on site (PHY-05)
POW, THM	Spares and lead-time strategy for electrical and cooling plant	Restoration requirements; maintenance plans
CMP, STO, NET	Firmware supply, compatibility and staged rollout	Firmware baselines and upgrade testing (CMP-04, NET-06)
OPS	Supplier service levels; vendor access approvals tied to tickets	Change and work control (OPS-02, OPS-05)
RES	Logistics and labour arrangements	Recovery scenarios and exercises (RES-03 to RES-05)
SEC	Third-party identity lifecycle; supplier advisories	Identity architecture, remote access, vulnerability management (SEC-01, SEC-03, SEC-04)
EVA	Evidence-rights mapping; supplier UNKNOWN Records	Evidence ownership and custody (EVA-01, EVA-02)

15 Informative references

Document-level references; no clause-level reliance is implied.

- ISO 9001 (quality management systems); ISO 44001 (collaborative business relationship management)
- ISO 22301 (business continuity management systems); ISO 22318 (supply chain continuity); ISO 28000 (supply chain security management)
- ISO/IEC 27001 (information security management systems); ISO/IEC 27036 series (information security for supplier relationships)
- ISO/IEC 20243 (Open Trusted Technology Provider Standard)
- NIST SP 800-161r1 (cyber supply chain risk management); NIST SP 800-218 (Secure Software Development Framework); NIST SP 800-193 (platform firmware resiliency); NIST SP 800-147 (BIOS protection); NIST SP 800-82r3 (OT security); NIST SP 800-53 Rev. 5 (AC and MA control families)
- IEC 62443 series, in particular IEC 62443-2-4 (service providers), IEC 62443-4-1 (secure product development) and IEC TR 62443-2-3 (patch management)
- ISO/IEC 29147 (vulnerability disclosure); ISO/IEC 30111 (vulnerability handling); FIRST CVSS; CISA Known Exploited Vulnerabilities catalogue
- ISO/IEC 5962 (SPDX); ECMA-424 (CycloneDX); NTIA minimum elements for an SBOM
- DMTF DSP0274 (SPDM); DMTF Redfish; IETF RFC 9334 (remote attestation architecture)

- SAE AS5553, SAE AS6081, SAE AS6171, SAE AS6174 (counterfeit parts avoidance, detection and mitigation)
- TAPA Freight Security Requirements
- IEC 62402 (obsolescence management); ISO 55001 (asset management systems); IEC 60300-3-11 (reliability centred maintenance)
- ISO/IEC 19086-1 (cloud SLA framework); ISO/IEC 20000-1 (service management)
- Directive (EU) 2022/2555 (NIS2) and Regulation (EU) 2022/2554 (DORA) — informative for in-scope sectors
- AIDCF-003, AIDCF-004, AIDCF-008, AIDCF-010, AIDCF-013, AIDCF-014

16 Control summary

Control	Title	Type	AL	Claim type	Gate mappings
MCF-SUP-001	Supplier Capability	PREVENTIVE	AL2	DESIGN	CG-03 (S)
MCF-SUP-002	Financial and Operational Resilience	DETECTIVE	AL4	OPERATION	CG-03 (S)
MCF-SUP-003	Security and Quality Posture	PREVENTIVE	AL3	IMPLEMENTATION	CG-08 (S); CG-15 (S)
MCF-SUP-004	Sub-Tier Visibility	PREVENTIVE	AL3	IMPLEMENTATION	CG-03 (C)
MCF-SUP-005	Service Levels	DIRECTIVE	AL3	IMPLEMENTATION	CG-10 (S); CG-13 (S)
MCF-SUP-006	Evidence Rights	DIRECTIVE	AL3	IMPLEMENTATION	CG-03 (S)
MCF-SUP-007	Incident Notification	DETECTIVE	AL4	OPERATION	CG-03 (S); CG-15 (S)
MCF-SUP-008	Exit and Transition Rights	DIRECTIVE	AL2	DESIGN	CG-10 (S)
MCF-SUP-009	Approved Sourcing	PREVENTIVE	AL3	IMPLEMENTATION	CG-08 (S); CG-04 (S)
MCF-SUP-010	Counterfeit Prevention	DETECTIVE	AL4	OPERATION	CG-04 (S); CG-08 (S)
MCF-SUP-011	Chain of Custody	PREVENTIVE	AL4	OPERATION	CG-08 (S)
MCF-SUP-012	Component Traceability	DETECTIVE	AL5	EFFECTIVENESS	CG-08 (S); CG-03 (S)
MCF-SUP-013	Release Provenance	PREVENTIVE	AL3	IMPLEMENTATION	CG-15 (S); CG-14 (S)
MCF-SUP-014	Signature Validation	PREVENTIVE	AL5	EFFECTIVENESS	CG-15 (S); CG-08 (S)
MCF-SUP-015	Compatibility Assurance	PREVENTIVE	AL5	EFFECTIVENESS	CG-14 (S); CG-08 (S); CG-09 (S)
MCF-SUP-016	Vulnerability Response	CORRECTIVE	AL4	OPERATION	CG-15 (S)
MCF-SUP-017	Critical Spares	RECOVERY	AL4	OPERATION	CG-10 (S); CG-05 (S); CG-06 (S)

Control	Title	Type	AL	Claim type	Gate mappings
MCF-SUP-018	Lead Times	DIRECTIVE	AL2	DESIGN	CG-10 (S); CG-03 (S)
MCF-SUP-019	Logistics Continuity	RECOVERY	AL5	RESILIENCE	CG-10 (S); CG-05 (S)
MCF-SUP-020	End-of-Life Planning	PREVENTIVE	AL4	OPERATION	CG-03 (S)
MCF-SUP-021	Support Identity	PREVENTIVE	AL3	IMPLEMENTATION	CG-15 (C); CG-13 (S)
MCF-SUP-022	Activation and Approval	PREVENTIVE	AL4	OPERATION	CG-15 (C); CG-14 (S)
MCF-SUP-023	Session Monitoring	DETECTIVE	AL4	OPERATION	CG-15 (S)
MCF-SUP-024	Revocation and Review	CORRECTIVE	AL4	OPERATION	CG-15 (S); CG-13 (S)

Key: (C) = CRITICAL role, (S) = SUPPORTING role, per AIDCF-010 gate roll-up. CG-11 and CG-12 are evaluated across all controls and are not mapped per control.

17 Worked synthetic example

> **Synthetic example.** All names, values and results below are invented for illustration. They do not describe any real supplier, product, facility or assessment.

Context. Synthetic Capability Instance CI-SYN-INF-02 is an inference platform in an owner-operated hall with direct-to-chip liquid cooling from six CDUs on a shared secondary loop, and a two-tier leaf-spine fabric. The decision question asks whether the platform can meet its availability outcome under NORMAL, MAINTENANCE and RECOVERY states.

Firmware rollout (MCF-SUP-015). The assessor reconstructed the last NIC firmware rollout. The compatibility matrix had been updated and a pre-production test passed. Rollout records showed three waves, each confined to one fabric pod, with a 24-hour soak and automated health gates; redundant members were never in one wave. Result: CONFORMING (EFFECTIVENESS, S4).

Signature validation (MCF-SUP-014). Verification logs covered all staged releases. Signature enforcement was enabled on BMCs and switches. The CDU controllers could not enforce signatures; this was recorded with a compensating hash check before upload. A rejection test had been performed on a lab BMC and lab switch within the interval. Result: CONFORMING.

Support identity (MCF-SUP-021). Account exports from the BMS, EPMS and two sampled CDU controllers found a generic "service" account on all six CDU controllers, shared by the CDU OEM's field engineers and not vaulted. Result: NONCONFORMING (Finding: shared vendor account on control-plane devices).

Activation and approval (MCF-SUP-022). The access gateway was off by default and all sampled IT sessions had approvals. However, the CDU OEM had a separate cellular modem on the CDU management network for remote diagnostics, always on and outside the gateway. Result: NONCONFORMING.

Critical spares (MCF-SUP-017). Strategy existed; physical count matched records; one spare CDU pump on site against six installed, with OEM regional stock committed at 48 hours. Result: CONFORMING. Supporting to CG-10.

Gate result. CG-15 has MCF-SUP-021 and MCF-SUP-022 as CRITICAL controls in this domain (with the SEC CRITICAL controls). Both are NONCONFORMING, so **CG-15 = FAIL** under the AIDCF-010 roll-up rule, regardless of the SEC results. CG-14 and CG-08 receive SUPPORTING inputs from MCF-SUP-015 (CONFORMING) and need no gate-level evaluation from this domain.

Decision effect and actions. A FAIL on CG-15 prevents a REASONABLY_ASSURED decision; the ceiling is that defined for CG-15 in AIDCF-010. Actions: remove the cellular modem or bring it behind the access gateway with off-by-default activation; replace the shared CDU account with named vendor identities or a vaulted per-use checkout; re-export accounts to verify. Reassessment trigger: closure of both Findings with E2 evidence.

What the example shows. The IT management plane was well controlled; the exposure sat in a facility controller supplied and supported by an OEM, reached through a path the IT team did not own. SUP-06 controls, applied to facility systems as well as IT, are what surface it.